Chapter 3

Plant Kingdom

Whittaker's five kingdoms were set out in the previous chapter. Here the classification within Kingdom Plantae, popularly known as the plant kingdom, is taken further.

What Plantae no longer holds

- our understanding of the plant kingdom has changed over time
- fungi, and the members of Monera and Protista that have cell walls, are now excluded, though earlier classifications placed them in the same kingdom
- so the cyanobacteria, also referred to as blue green algae, are not algae any more

Artificial systems

- the earliest systems used only gross superficial morphological characters such as habit, colour, number and shape of leaves
- they rested mainly on vegetative characters, or on the androecium structure in the system given by Linnaeus
- being based on a few characters they separated closely related species, and they gave equal weight to vegetative and sexual characters, which is not acceptable since vegetative characters are more easily affected by environment

Natural systems

- developed against the artificial ones, and based on natural affinities among organisms
- they consider not only external features but internal ones too: ultrastructure, anatomy, embryology and phytochemistry

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3.2 Algae

3.5 Bryophytes

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3.1 How flowering plants came to be classified

Natural and phylogenetic systems

- a natural classification for flowering plants was given by George Bentham and Joseph Dalton Hooker
- at present phylogenetic systems, based on evolutionary relationships, are acceptable; they assume that organisms of the same taxa have a common ancestor
- information from many other sources helps resolve difficulties, and matters more where there is no supporting fossil evidence

Three modern aids

- numerical taxonomy, now easily done on computers, is based on all observable characters; numbers and codes are assigned to every character and the data processed, so each character carries equal importance and hundreds can be considered at once
- cytotaxonomy rests on cytological information such as chromosome number, structure and behaviour
- chemotaxonomy uses the chemical constituents of the plant to resolve confusions

3.2 Algae

What algae are

- chlorophyll-bearing, simple, thalloid, autotrophic and largely aquatic, in both fresh water and marine habitats
- they occur in a variety of other habitats too: moist stones, soils and wood
- some occur in association with fungi as lichen, and with animals, for instance on sloth bear
- form and size are highly variable, from colonial forms such as Volvox to filamentous forms such as Ulothrix and Spirogyra, while a few marine forms such as kelps make massive bodies

How algae reproduce

- vegetative reproduction is by fragmentation, each fragment developing into a thallus
- asexual reproduction is by different types of spores, most commonly the flagellated motile zoospores, which on germination give rise to new plants
- sexual reproduction is by fusion of two gametes: isogamous where the gametes are similar in size, whether flagellated as in Ulothrix or non-motile as in Spirogyra
- anisogamous where the two differ in size, as in species of Eudorina, and oogamous where a large non-motile female gamete fuses with a smaller motile male one, as in Volvox and Fucus

Green algae

- Volvox, a colonial form
- Ulothrix, a filamentous form

Brown algae

- Laminaria
- Fucus
- Dictyota

Red algae

- Porphyra
- Polysiphonia

Figure 3.A Algae : (a) Green algae (i) Volvox (ii) Ulothrix (b) Brown algae (i) Laminaria (ii) Fucus (iii) Dictyota (c) Red algae (i) Porphyra (ii) Polysiphonia

How algae are useful

- at least half of the total carbon dioxide fixation on earth is carried out by algae through photosynthesis, and being photosynthetic they raise dissolved oxygen in their surroundings
- they are of paramount importance as primary producers of energy-rich compounds, which form the basis of the food cycles of all aquatic animals
- many species of Porphyra, Laminaria and Sargassum are among the 70 species of marine algae used as food, and Chlorella, a unicellular alga rich in proteins, is a food supplement even for space travellers
- certain marine brown and red algae yield large amounts of hydrocolloids, water holding substances: algin from brown algae and carrageen from red algae; agar from Gelidium and Gracilaria is used to grow microbes and in ice-creams and jellies

3.3 Chlorophyceae and phaeophyceae

Chlorophyceae, the green algae

- the plant body may be unicellular, colonial or filamentous, and is usually grass green from the dominance of chlorophyll a and b, localised in definite chloroplasts
- chloroplasts may be discoid, plate-like, reticulate, cup-shaped, spiral or ribbon-shaped
- most have one or more storage bodies called pyrenoids in the chloroplasts, holding protein besides starch; some store food as oil droplets
- the rigid cell wall has an inner layer of cellulose and an outer layer of pectose
- reproduction is vegetative by fragmentation, asexual by flagellated zoospores in zoosporangia, and sexual with isogamous, anisogamous or oogamous forms; examples are Chlamydomonas, Volvox, Ulothrix, Spirogyra and Chara

Phaeophyceae, the brown algae

- found primarily in marine habitats, with great variation in size and form
- they range from simple branched filamentous forms such as Ectocarpus to the profusely branched kelps, which may reach a height of 100 metres
- they possess chlorophyll a and c, carotenoids and xanthophylls
- colour varies from olive green to shades of brown depending on how much of the xanthophyll pigment fucoxanthin is present

The brown algal body

- food is stored as complex carbohydrates, which may be laminarin or mannitol
- vegetative cells have a cellulosic wall usually covered outside by a gelatinous coating of algin; the protoplast holds plastids, a centrally located vacuole and a nucleus
- the body is usually attached to the substratum by a holdfast, and has a stalk, the stipe, and a leaf like photosynthetic organ, the frond

Brown algal reproduction

- vegetative reproduction is by fragmentation
- asexual reproduction in most is by biflagellate zoospores, pear-shaped, with two unequal laterally attached flagella
- sexual reproduction may be isogamous, anisogamous or oogamous, the union taking place in water or within the oogonium in oogamous species
- gametes are pyriform, that is pear-shaped, and bear two laterally attached flagella; common forms are Ectocarpus, Dictyota, Laminaria, Sargassum and Fucus

3.4 Rhodophyceae

The red algae

- called red algae from the predominance of the red pigment r-phycoerythrin in the body
- the majority are marine, in greater concentrations in warmer areas, occurring both in well-lighted regions close to the surface and at great depths where little light penetrates
- the red thalli of most are multicellular and some have complex body organisation
- food is stored as floridean starch, very similar in structure to amylopectin and glycogen
- they reproduce vegetatively by fragmentation and asexually by non-motile spores

Table 3.A Divisions of Algae and their Main Characteristics

Classes	Common Name	Major Pigments	Stored Food	Cell Wall	Flagellar Number and Position of Insertions	Habitat
Chlorophyceae	Green algae	Chlorophyll a, b	Starch	Cellulose	2-8, equal, apical	Fresh water, brackish water, salt water
Phaeophyceae	Brown algae	Chlorophyll a, c, fucoxanthin	Mannitol, laminarin	Cellulose and algin	2, unequal, lateral	Fresh water (rare), brackish water, salt water
Rhodophyceae	Red algae	Chlorophyll a, d, phycoerythrin	Floridean starch	Cellulose, pectin and poly sulphate esters	Absent	Fresh water (some), brackish water, salt water (most)

Marchantia, a liverwort

- female thallus, bearing the archegoniophore
- male thallus, bearing the antheridiophore
- gemma cups on the thallus surface
- rhizoids anchoring the thallus below

Funaria and Sphagnum, mosses

- main axis with leaves, the gametophyte
- antheridial branch and archegonial branch
- sporophyte borne on the gametophyte
- seta carrying the capsule at its tip

Figure 3.B Bryophytes: A liverwort – Marchantia (a) Female thallus (b) Male thallus Mosses – (c) Funaria, gametophyte and sporophyte (d) Sphagnum gametophyte

Red algae, concluded

- sexual reproduction is by non-motile gametes and is oogamous
- it is accompanied by complex post fertilisation developments
- common members are Polysiphonia, Porphyra, Gracilaria and Gelidium

Bryophytes

- the various mosses and liverworts
- found commonly growing in moist shaded areas in the hills

3.5 Bryophytes

Amphibians of the plant kingdom

- so called because they can live in soil but depend on water for sexual reproduction
- they usually occur in damp, humid and shaded localities
- they play an important role in plant succession on bare rocks and soil

The bryophyte body

- more differentiated than that of algae: thallus-like, prostrate or erect, attached to the substratum by unicellular or multicellular rhizoids
- they lack true roots, stem or leaves, though they may possess root-like, leaf-like or stem-like structures
- the main plant body is haploid and produces gametes, hence is called a gametophyte

Sex organs and the sporophyte

- the sex organs are multicellular; the male organ, the antheridium, produces biflagellate antherozoids, and the female organ, the archegonium, is flask-shaped and produces a single egg
- antherozoids are released into water, where they reach the archegonium, and one fuses with the egg to give the zygote
- zygotes do not undergo reduction division immediately; they produce a multicellular sporophyte, which is not free-living but attached to the photosynthetic gametophyte and draws nourishment from it
- some cells of the sporophyte undergo meiosis to produce haploid spores, which germinate to produce the gametophyte

Why bryophytes matter

- in general of little economic importance, though some mosses feed herbaceous mammals, birds and other animals
- species of Sphagnum provide peat, long used as fuel and as packing material for shipping living material, because of its capacity to hold water
- mosses with lichens are the first to colonise rocks, decomposing them and making the substrate fit for higher plants
- forming dense mats on soil, they reduce the impact of falling rain and prevent soil erosion

Liverworts

- grow in moist shady habitats: banks of streams, marshy ground, damp soil, bark of trees and deep in the woods
- the plant body is thalloid, as in Marchantia, the thallus dorsiventral and closely appressed to the substrate; leafy members carry tiny leaf-like appendages in two rows
- asexual reproduction is by fragmentation of thalli, or by gemmae, green multicellular asexual buds developing in receptacles called gemma cups, which detach and germinate into new individuals

3.6 Mosses

Liverworts, concluded

- in sexual reproduction the male and female sex organs are produced either on the same thallus or on different ones
- the sporophyte is differentiated into a foot, seta and capsule
- after meiosis spores are produced within the capsule, and these germinate to form free-living gametophytes

The moss gametophyte

- the predominant stage of the life cycle, and it runs in two stages
- the protonema stage develops directly from a spore: creeping, green, branched and frequently filamentous
- the leafy stage develops from the secondary protonema as a lateral bud, with upright slender axes bearing spirally arranged leaves, attached to soil by multicellular branched rhizoids
- the leafy stage bears the sex organs

Moss reproduction

- vegetative reproduction is by fragmentation and by budding in the secondary protonema
- in sexual reproduction the antheridia and archegonia are produced at the apex of the leafy shoots
- after fertilisation the zygote develops into a sporophyte of foot, seta and capsule, more elaborate than that of liverworts; the capsule holds spores formed after meiosis
- mosses have an elaborate mechanism of spore dispersal; common examples are Funaria, Polytrichum and Sphagnum

3.7 Pteridophytes

Horsetails and ferns

- used for medicinal purposes and as soil-binders, and frequently grown as ornamentals
- evolutionarily they are the first terrestrial plants to possess vascular tissues, xylem and phloem
- they are found in cool, damp, shady places, though some flourish in sandy-soil conditions

A sporophyte-dominant body

- unlike bryophytes, the main plant body is a sporophyte differentiated into true root, stem and leaves, with well-differentiated vascular tissues
- leaves are small microphylls, as in Selaginella, or large macrophylls, as in ferns
- sporophytes bear sporangia subtended by leaf-like appendages called sporophylls, which in some cases form compact structures called strobili or cones, as in Selaginella and Equisetum
- the sporangia produce spores by meiosis in spore mother cells

Selaginella and Equisetum

- Selaginella, with small leaves, the microphylls
- Equisetum, showing the strobilus at the shoot tip
- node and internode marked along the axis
- branches arising in whorls at the nodes

Fern and Salvinia

- Fern, with large leaves, the macrophylls
- the rhizome running below the ground
- Salvinia, one of the heterosporous genera

Figure 3.C Pteridophytes : (a) Selaginella (b) Equisetum (c) Fern (d) Salvinia

The prothallus

- spores germinate into inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus
- these need cool, damp, shady places to grow, and that restricted requirement, with the need for water at fertilisation, limits living pteridophytes to narrow geographical regions
- the gametophytes bear antheridia and archegonia; water carries the antherozoids released from the antheridia to the mouth of the archegonium, and fusion with the egg gives the zygote
- the zygote produces a multicellular well-differentiated sporophyte, the dominant phase

Homospory and heterospory

- in the majority of pteridophytes all spores are of one kind, and such plants are homosporous
- genera such as Selaginella and Salvinia produce two kinds, large megaspores and small microspores, and are heterosporous
- megaspores and microspores germinate into female and male gametophytes respectively, and the female gametophytes are retained on the parent sporophyte for variable periods
- the zygote develops into a young embryo within the female gametophyte, an event that is a precursor to the seed habit and an important step in evolution

Four classes

- Psilopsida, with Psilotum
- Lycopsida, with Selaginella and Lycopodium
- Sphenopsida, with Equisetum
- Pteropsida, with Dryopteris, Pteris and Adiantum

3.8 Gymnosperms

Naked seeds

- gymnos means naked and sperma means seeds: the ovules are not enclosed by any ovary wall and stay exposed both before and after fertilisation, so the seeds that develop are naked
- they include medium-sized or tall trees and shrubs; the giant redwood Sequoia is one of the tallest tree species
- roots are generally tap roots; some genera carry a fungal association as mycorrhiza, in Pinus, while in Cycas small specialised coralloid roots associate with nitrogen-fixing cyanobacteria

Stems and leaves

- stems are unbranched in Cycas or branched in Pinus and Cedrus
- leaves may be simple or compound, and the pinnate leaves of Cycas persist for a few years
- the leaves are well-adapted to withstand extremes of temperature, humidity and wind
- in conifers the needle-like leaves reduce surface area, and a thick cuticle with sunken stomata further reduces water loss

Cycas, with unbranched stem and pinnate leaves

Pinus, showing dwarf shoot and long shoot

Seeds borne exposed on the cone

Ginkgo, the third genus shown

Figure 3.D Gymnosperms: (a) Cycas (b) Pinus (c) Ginkgo

3.9 Cones, spores and seeds

Two kinds of spore, two kinds of cone

- gymnosperms are heterosporous, producing haploid microspores and megaspores within sporangia borne on sporophylls, which are arranged spirally along an axis to form lax or compact strobili or cones
- strobili bearing microsporophylls and microsporangia are microsporangiate or male strobili
- cones bearing megasporophylls with ovules or megasporangia are macrosporangiate or female strobili
- male and female cones may be borne on the same tree, as in Pinus, but in Cycas the male cones and megasporophylls are borne on different trees

The pollen grain and the ovule

- microspores develop into a highly reduced male gametophytic generation confined to a limited number of cells, and this reduced gametophyte is the pollen grain, developing within the microsporangium
- the megaspore mother cell differentiates from one of the cells of the nucellus, which is protected by envelopes, the composite structure being the ovule
- the megaspore mother cell divides meiotically into four megaspores, and one of them, enclosed within the megasporangium, develops into a multicellular female gametophyte bearing two or more archegonia, retained within the megasporangium

Fertilisation

- unlike in bryophytes and pteridophytes, neither gametophyte has an independent free-living existence; both remain within the sporangia retained on the sporophyte
- the pollen grain is released from the microsporangium and is carried in air currents to the opening of the ovules borne on megasporophylls
- the pollen tube carrying the male gametes grows towards the archegonia and discharges its contents near their mouth
- after fertilisation the zygote develops into an embryo and the ovules into seeds, which are not covered

A dicotyledon, one of the two classes

A monocotyledon, the other class

In both, pollen grains and ovules develop in flowers

In both, the seeds are enclosed in fruits

Figure 3.E Angiosperms : (a) A dicotyledon (b) A monocotyledon

3.10 Angiosperms

Flowering plants

- unlike gymnosperms, where the ovules are naked, in angiosperms the pollen grains and ovules develop in specialised structures called flowers, and the seeds are enclosed in fruits
- an exceptionally large group occurring in a wide range of habitats
- they range in size from the smallest Wolffia to tall trees of Eucalyptus, over 100 metres
- they provide food, fodder, fuel, medicines and several other commercially important products

3.11 Summary

Algae

- the plant kingdom takes in algae, bryophytes, pteridophytes, gymnosperms and angiosperms
- algae are chlorophyll-bearing simple, thalloid, autotrophic and largely aquatic organisms
- by pigment and stored food they fall into Chlorophyceae, Phaeophyceae and Rhodophyceae
- they reproduce vegetatively by fragmentation, asexually by spores and sexually by gametes, showing isogamy, anisogamy or oogamy

Bryophytes

- they can live in soil but depend on water for sexual reproduction
- the body is more differentiated than that of algae: thallus-like, prostrate or erect, attached by rhizoids, with root-like, leaf-like and stem-like structures

Bryophytes and pteridophytes, concluded

- bryophytes divide into liverworts and mosses; the liverwort body is thalloid and dorsiventral, while mosses have upright slender axes bearing spirally arranged leaves
- the main bryophyte body is gamete-producing, the gametophyte, bearing antheridia and archegonia; the gametes fuse to a zygote that produces a multicellular sporophyte, which produces haploid spores that germinate to form gametophytes
- in pteridophytes the main plant is a sporophyte of true root, stem and leaves with well-differentiated vascular tissues, bearing sporangia that produce spores
- those spores form gametophytes needing cool damp places, bearing antheridia and archegonia; water is required to carry male gametes to the archegonium, where the zygote forms

Gymnosperms and angiosperms, concluded

- in gymnosperms the ovules are not enclosed by any ovary wall, and after fertilisation the seeds remain exposed, so these are called naked-seeded plants
- they produce microspores and megaspores in microsporangia and megasporangia borne on sporophylls, and the microsporophylls and megasporophylls are arranged spirally on an axis to form male and female cones
- the pollen grain germinates and the pollen tube releases the male gamete into the ovule, where it fuses with the egg cell in the archegonia; the zygote develops into an embryo and the ovules into seeds
- the angiosperms are divided into two classes, the dicotyledons and the monocotyledons

3.12 Exercises

From this chapter

1. What is the basis of classification of algae?
2. When and where does reduction division take place in the life cycle of a liverwort, a moss, a fern, a gymnosperm and an angiosperm?
3. Name three groups of plants that bear archegonia. Briefly describe the life cycle of any one of them.
4. Mention the ploidy of the following: protonemal cell of a moss; primary endosperm nucleus in dicot; leaf cell of a moss; prothallus cell of a fern; gemma cell in Marchantia; meristem cell of monocot; ovum of a liverwort; and zygote of a fern.

From this chapter, continued

1. Write a note on economic importance of algae and gymnosperms.
2. Both gymnosperms and angiosperms bear seeds, then why are they classified separately?
3. What is heterospory? Briefly comment on its significance. Give two examples.
4. Explain briefly the following terms with suitable examples: (i) protonema (ii) antheridium (iii) archegonium (iv) diplontic (v) sporophyll (vi) isogamy
5. Differentiate between the following: (i) red algae and brown algae (ii) liverworts and moss (iii) homosporous and heterosporous pteridophyte
6. Match column I with column II: (a) Chlamydomonas (b) Cycas (c) Selaginella (d) Sphagnum, against (i) Moss (ii) Pteridophyte (iii) Algae (iv) Gymnosperm
7. Describe the important characteristics of gymnosperms.